

Itô's Lemma, Martingales, and Their Role in Stochastic Calculus

Your Name

May 21, 2025

Introduction

This document synthesizes key concepts in stochastic calculus:

- Itô processes and Itô's Lemma,
- The martingale property,
- Connections to PDEs (Feynman-Kac),
- Applications in finance and physics.

1 Itô Processes and Itô's Lemma

Definition of an Itô Process

An **Itô process** X_t solves the SDE:

$$dX_t = \mu(t, X_t)dt + \sigma(t, X_t)dB_t,$$

where:

- $\mu(t, X_t)$: Drift (deterministic),
- $\sigma(t, X_t)$: Diffusion (volatility),
- B_t : Standard Brownian motion.

Itô's Lemma (1D Case)

For a $C^{1,2}$ function $u(t, x)$, the stochastic differential of $u(t, X_t)$ is:

$$du = \underbrace{\left(\frac{\partial u}{\partial t} + \mu \frac{\partial u}{\partial x} + \frac{\sigma^2}{2} \frac{\partial^2 u}{\partial x^2} \right) dt}_{\text{Drift}} + \underbrace{\sigma \frac{\partial u}{\partial x} dB_t}_{\text{Noise}}.$$

Why the Extra Term?

Brownian motion has **quadratic variation** $[B]_t = t$, leading to $dB_t^2 = dt$. The Taylor expansion of du must include:

$$(dX_t)^2 = \sigma^2 dB_t^2 + (\text{small terms}) = \sigma^2 dt.$$

Example: Geometric Brownian Motion

For $dS_t = \mu S_t dt + \sigma S_t dB_t$, apply Itô's Lemma to $u(S_t) = \ln S_t$:

$$d(\ln S_t) = \left(\mu - \frac{\sigma^2}{2} \right) dt + \sigma dB_t.$$

This solves to:

$$S_t = S_0 e^{(\mu - \frac{\sigma^2}{2})t + \sigma B_t}.$$

2 Martingales

Definition

A process M_t is a **martingale** if:

$$\mathbb{E}[M_t \mid \mathcal{F}_s] = M_s \quad \text{for all } s \leq t,$$

where $\mathcal{F}_s$ is the filtration (information up to time s).

Key Properties

- **No drift:** Martingales model "fair games" (no net trend).
- **Example:** Brownian motion B_t is a martingale.
- **Itô processes:** If $dX_t = \sigma(t, X_t)dB_t$ (no drift), X_t is a martingale.

Connection to Itô's Lemma

For $u(t, X_t)$ to be a martingale:

$$\text{Drift term} = 0 \implies \frac{\partial u}{\partial t} + \mu \frac{\partial u}{\partial x} + \frac{\sigma^2}{2} \frac{\partial^2 u}{\partial x^2} = 0.$$

This is the **Kolmogorov backward equation**.

3 Feynman-Kac Formula

Statement

The solution to the PDE:

$$\frac{\partial u}{\partial t} + \mu \frac{\partial u}{\partial x} + \frac{\sigma^2}{2} \frac{\partial^2 u}{\partial x^2} = 0, \quad u(T, x) = \psi(x),$$

is given by:

$$u(t, x) = \mathbb{E}[\psi(X_T) \mid X_t = x],$$

where X_t follows $dX_t = \mu dt + \sigma dB_t$.

Interpretation

PDE solutions can be expressed as expectations of Itô processes.

4 Applications

Black-Scholes PDE

The option price $V(t, S_t)$ under geometric Brownian motion solves:

$$\frac{\partial V}{\partial t} + rS \frac{\partial V}{\partial S} + \frac{\sigma^2 S^2}{2} \frac{\partial^2 V}{\partial S^2} - rV = 0.$$

By Feynman-Kac:

$$V(t, S_t) = \mathbb{E}^\mathbb{Q}[e^{-r(T-t)} \psi(S_T) \mid S_t].$$

Dirichlet Problem

The solution to $\Delta u = 0$ in a domain D is:

$$u(x) = \mathbb{E}[\phi(B_\tau) \mid B_0 = x],$$

where τ is the first exit time of B_t from D .

Summary

- **Itô's Lemma:** Stochastic chain rule with a second-order term.
- **Martingales:** Processes with no drift, central to pricing.
- **Feynman-Kac:** Bridges SDEs and PDEs via expectations.